DBSgram User Guide

Local Field Potentials and Hand Motor Performance Data analysis Toolbox in the context of Deep Brain Stimulation parametrization

This document was created to serve as a user manual for clinicians and researchers alike, in order to allow them to take the most relevant insights out of the data collected, when using this toolbox.

The toolbox was designed with the aim of helping the users build visual reports for amplitude titration of Deep Brain Stimulation (DBS) therapies in Parkinson's Disease (PD) patients. The authors coined these reports as DBSgrams. It was implemented for kinematic data analysis in the context of 3 relevant hand motor performance tasks assessments, namely, resting hand tremor, wrist flexion-extension maneuvers and hand open-close. These evaluate part of the 3 cardinal motor symptoms in PD, respectively, tremor, rigidity and bradykinesia.

The toolbox is available on GitHub (https://github.com/NCN-Lab/DBSgram) as both a source code downloadable .zip file and a released standalone application. Therefore, this guide will explain first how to set up properly Matlab, for the source code version, or the application for the best use of the toolbox and define the minimum requirements to make it work. Afterwards, the sequential data processing pipeline is briefly explained, with the provided figures allowing the user to confirm if their outputs match the intended ones. Finishing all the inputs and processing of data streams, enables the user to generate the PDF report of the DBSgram.

If any issues arise from the use of this toolbox, please contact pauloaguiar.i3s.up.pt by e-mail with the print of the error message attached and your description/log of what led to that error. The authors would be glad to make this tool more aligned with their users' needs.

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Setting up the data before starting

In this section, details on how the data should be organized in order to be correctly parsed by the toolbox are presented. Before initiating the toolbox, ensure the patient recording directory contains both the native Medtronic Percept JSON export and the complete set of wearable sensor CSV files. For the UPDRS scores to be parsed automatically, ensure that the directory also has a UPDRS.xlsx, as the one provided for each example in the data folder, however, this is not a requirement and this data can be input manually.

The CSV files naming should follow a strict four-token underscore-delimited format to index the wearable recordings:

$$\langle \mathbf{ProtocolIndex} \rangle_\langle \mathbf{SensorType} \rangle_\langle \mathbf{SymptomTask} \rangle_\langle \mathbf{BodySide} \rangle . \mathbf{csv}$$

- ⟨**ProtocolIndex**⟩: Numeric sequence identifier matching the testing block (0, 1, 2, 3, etc.).
- ⟨**SensorType**⟩: Hardware mounting point: FingerSensor (proximal phalanx) or HandSensor (wrist/palm of the hand).
- ⟨**SymptomTask**⟩: Evaluated motor domain:
  - RestingTremor → automatically normalized to **Tremor** in workspace memory.
  - WristRigidity → automatically normalized to **Rigidity** in workspace memory.
  - OpenCloseHand → automatically normalized to **Bradykinesia** in workspace memory.
- ⟨**BodySide**⟩: Evaluated anatomical side (L or R).

After verifying the rules for file naming, make sure also that every testing block index contains exactly 4 CSV files (FingerSensor_L, FingerSensor_R, HandSensor_L, HandSensor_R), since the data ingestion engine files into sequential 4-channel packs; omitting a sensor desynchronizes subsequent task groupings.

Additionally, the CSV should include a metadata header containing the line Test_starttime=YYYY-MM-DD HH:mm:ss.SSSSSSZ. The toolbox extracts this timestamp to establish the wearable clock baseline; if missing or corrupted, chronological sorting and initial IPG offset calculations will fail. Sensor data rows must begin with an ISO-style timestamp (YYYY-MM-DD HH:mm:ss.SSS) followed by 6 IMU telemetry columns in standard units:

$$[\mathbf{time},\ \mathbf{ax},\ \mathbf{ay},\ \mathbf{az},\ \mathbf{gx},\ \mathbf{gy},\ \mathbf{gz}]$$

where acceleration is in g and angular velocity is in deg/s. The order of these inputs must be respected. Any further data entries on the same line will be ignored.

Please also make sure that only one .json file exists in the directory. The parser will default to load the first file alphabetically, if multiple files are present.

Matlab Setup

If the user is using the standalone app version of this toolbox, they can skip this section, since the application already had dependencies handled on creation. For the users resorting to source code download, please ensure beforehand that the following is installed:

- Matlab version should be 2025a (or more recent versions)
- DSP System Toolbox
- Optimization Toolbox
- Signal Processing Toolbox
- Statistics and Machine Learning Toolbox
- Wavelet Toolbox

Additionally, the user needs to set the appropriate path to enable the use of the toolbox. After opening Matlab, search for the ***Home*** Tab and on the ***Environment Panel***, select ***Set Path*** (see Fig.1). On the window it opens, select ***Add Folder >> Add Folder to Path***. This opens a file explorer window, in which you should look for the unzipped folder with the source code and then press ***OK*** at the bottom right of the window.

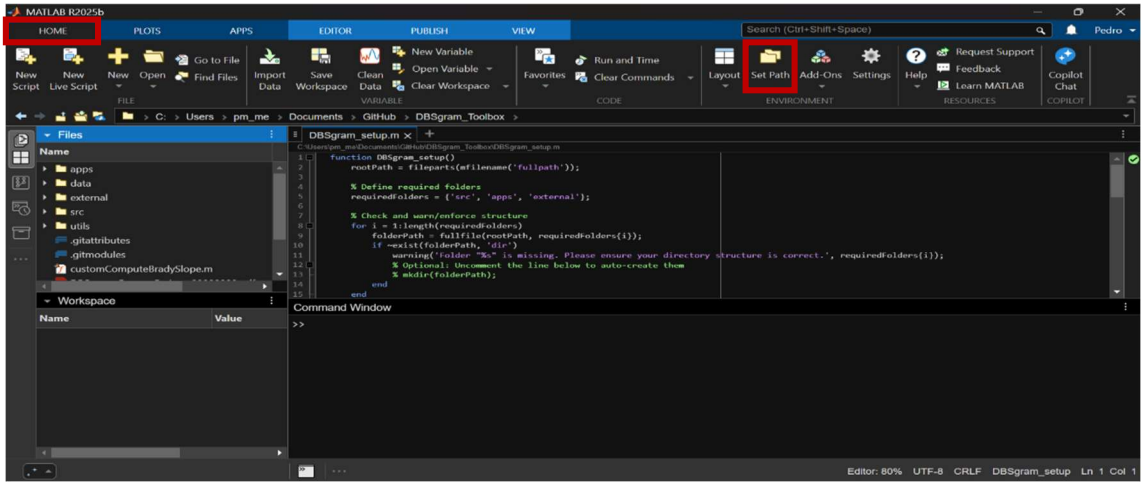


Figure 1 – MATLAB main UI in the Home tab menu.

Finally, double-check that the ***DBSgram_Toolbox >> external >> DBScope*** folder is not empty. If you downloaded the toolbox as a .zip file, the ***DBScope*** subfolder will be empty by default; go to https://github.com/NCN-Lab/DBScope, download the .zip and extract the contents into the ***external >> DBScope*** folder, ensuring the subfolder is named "DBScope" and not "DBScope-main". If you are cloning via Git, you can bypass this manual step by cloning the repository with the recursive flag (see how on the following link https://github.com/pmtmelo/DBSgram_Toolbox.git). Once the path and ***DBScope*** folder are ready, initialize and launch the toolbox by typing *DBSgram_MainUI* into the MATLAB Command Window and pressing *Enter*.

Toolbox Initialization

To initialize the toolbox, write on the command window *DBSgram_MainUI*. This will run some checks on the accessible functions and the DBScope Integration. It also launches the **DBSgram Titration Preprocessing & Reporting Framework Suite** window.

The **DBSgram Titration Preprocessing & Reporting Framework Suite** operates as a structured, sequential pipeline divided into three main operational panels:

1. Data Loading and Temporal Alignment
2. Data Pre-Processing and Quality Audit
3. Clinical Metrics & Reports Generation

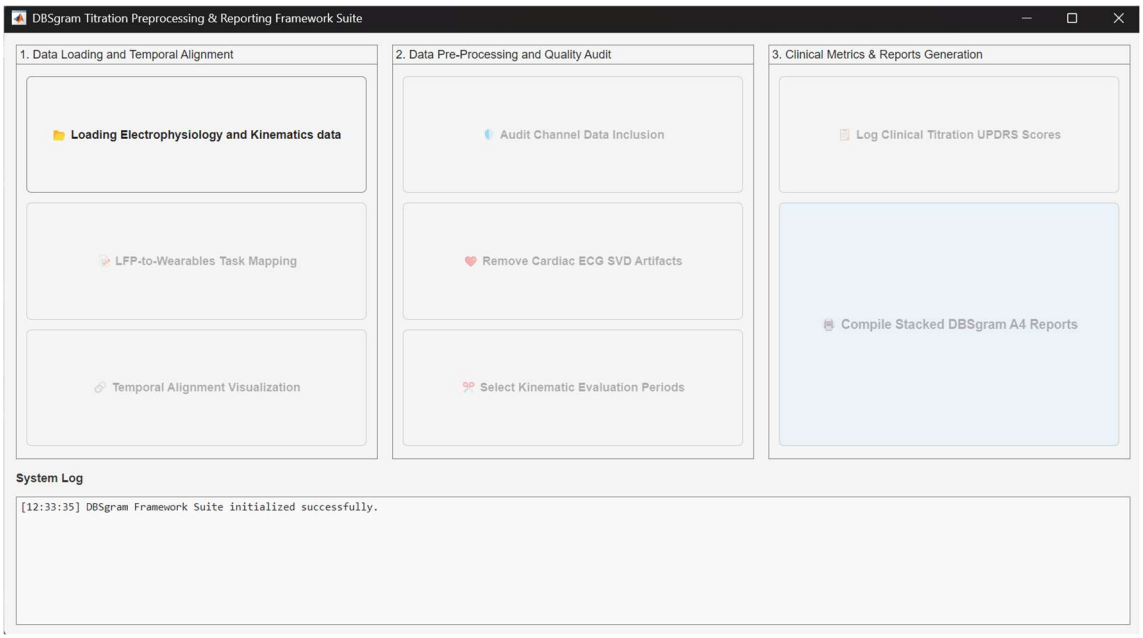


Figure 2 – Main Framework Suite initial state with only data loading enabled

Note: Button access is strictly gated: downstream modules remain locked until prerequisite data ingestion, alignment, and verification milestones are completed and validated in memory.

Step-by-Step Operational Workflow

1) Ingestion of Electrophysiological and Kinematic Data

On the main interface (see Fig. 2), locate *Panel 1 (Data Loading and Temporal Alignment)* and click the **Loading Electrophysiology and Kinematics data** button. A native file explorer dialog will appear titled *"Select the folder containing BOTH the LFP .json and Wearables .csv files".*

Navigate to the session directory containing the patient's JSON export alongside all synchronized wearable IMU CSV data streams. Data ingestion assumes that the electrophysiology stream is the raw JSON file retrieved from the Medtronic's Percept Neurostimulators and that wearables files follow the rules in the previous section.

The framework automatically parses the streams, resolving sample clocks and repairing any transmission packet dropouts. During this ingestion phase, the engine automatically performs a preliminary matching before manual inspection:

- Duration Filtering**:** LFP recordings shorter than 45 seconds are flagged as telemetry checks and filtered out.

- Stimulation-Aware Task Classification**:** The engine inspects current amplitude vectors across both hemispheres to propose protocol IDs:
    - Assigns the initial runs to Task 0 (Tapping Sync) and Task 1 (Resting Baseline).
    - Detects which hemisphere modulates stimulation current while the contralateral side remains static, pre-labeling titration tasks sequentially (e.g., 2L, 3L, 4L vs. 2R, 3R, 4R)
    - Identifies sustained bilateral constant stimulation at the end of the session as Task 5.
- Wearable Pack Duration Matching: Wearable files are bundled into 4-sensor packs (bilateral hand and finger sensors). Each pack is paired to the candidate LFP recording with the closest matching elapsed duration within an adaptive tolerance window (< 20% duration difference or < 40 s).

The System Log at the bottom of the main suite will output the parsing progress, file matching diagnostics, and packet durations. Upon completion, the **Loading Electrophysiology and Kinematics data** button turns Green, and the **LFP-to-Wearables Task Mapping** button becomes active (see Fig. 3)

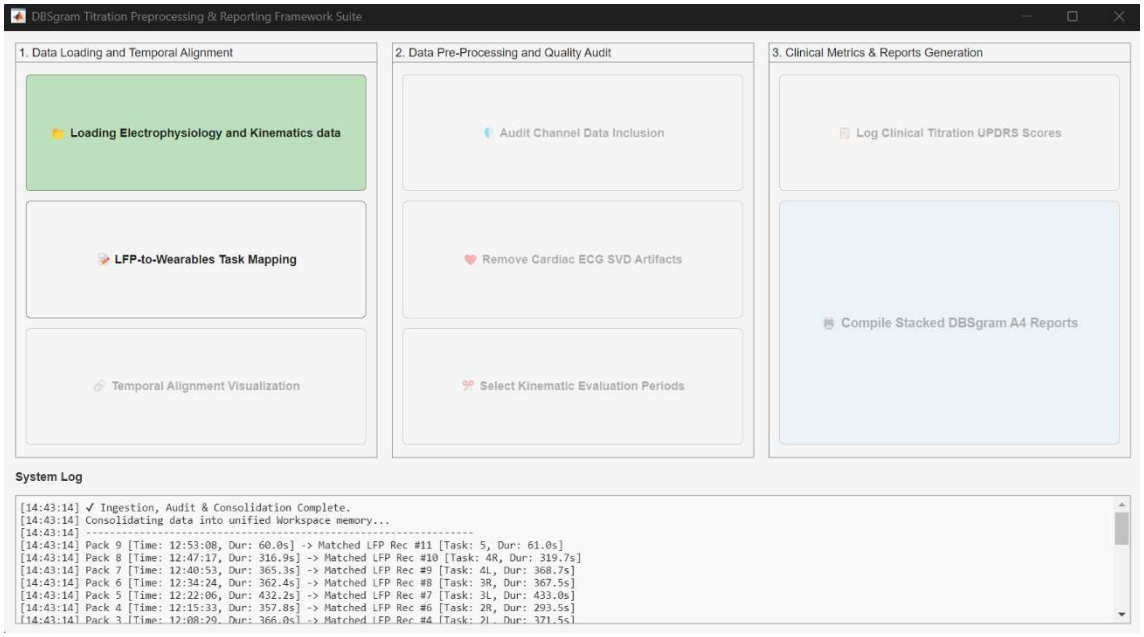


Figure 3 - Framework Suite following successful ingestion and consolidation

2) LFP-to-Wearables Task Mapping

Because clinical recordings frequently contain interrupted passes or brief aborted recordings, each electrophysiological telemetry stream must be mapped to its corresponding motor evaluation protocol. To that end, the user should click on the **LFP-to-Wearables Task Mapping** button, displaying a GUI which allows the user to audit the detected LFP files, as it can be seen in Fig. 4:

- LFP Rec #: Sequential index of the raw recording file.
- Duration: Total elapsed time of the recording in seconds.

- Stable Stimulation Levels (Hemisphere Breakdown): Identified current amplitudes (mA) and their respective hold times per hemisphere.
- Mapped Task ID: Dropdown selector for protocol assignment.

The user can then assign the appropriate Task ID using the dropdown for every row:

- <Discard>: Assigned to aborted sweeps or too short recordings
- 0: Synchronization & Baseline protocol (initial tapping alignment run)
- 1: Baseline calibration and rest electrophysiology profile.
- 2L/2R: Tremor clinical evaluation (Left/Right hemisphere target)
- 3L/3R: Rigidity sweep protocol (Left/Right hemisphere target)
- 4L/4R: Bradykinesia assessments through the hand open-close task (Left/Right hemisphere target)
- 5: Baseline at clinical setting stimulation, as set at patient came to the clinic.



Figure 4 - LFP to Wearables Task Mapping configuration table

Once recordings are appropriately assigned, the user should click the **Save Mappings & Close** button at the bottom. In the background, <Discard> data is effectively discarded and the wearables and LFP data is matched. The main GUI should then look like Fig. 5.

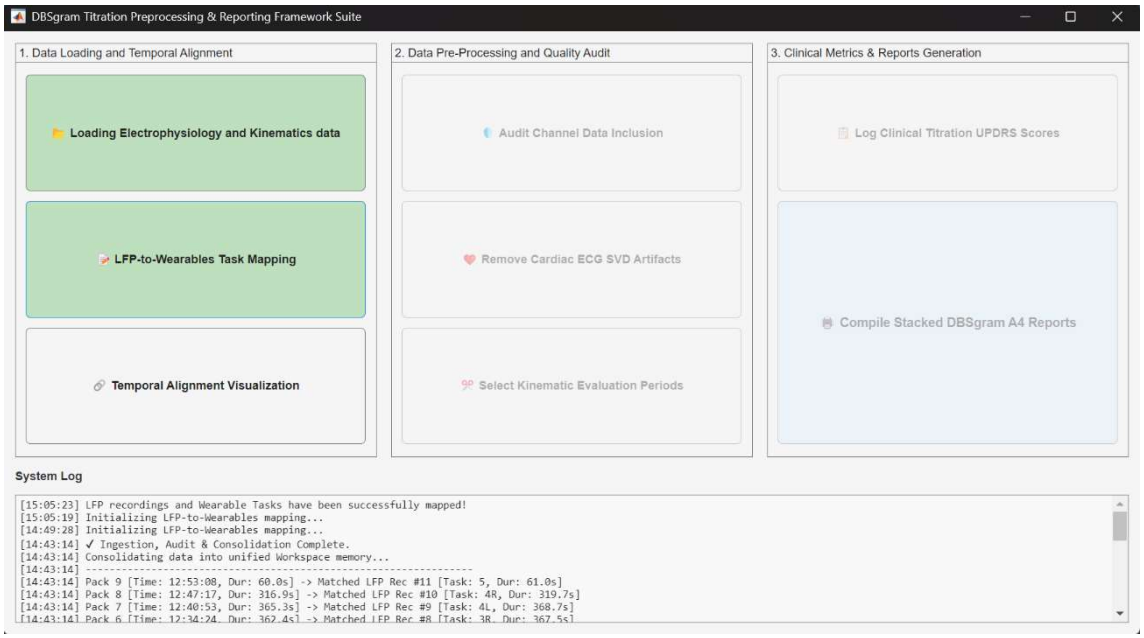


Figure 5 - Main Framework Suite with Temporal Alignment Visualization unlocked

3) Sub-Second Temporal Alignment Visualization

To bridge the clock offset between the neurostimulator and the smartphone-synchronized wearables data, an interactive micro-alignment step is performed using the initial tapping block (Task 0). A new dedicated interface is open when the user clicks on **Temporal Alignment Visualization** called *DBSgram Sync Alignment Workspace*.

The interface (Fig. 6) displays three stacked, synchronized axes:

- Top trace (blue): Target neural LFP filtered in the high-frequency artifact band (35-60 Hz), showing the electrical contact transients caused by physical tapping.
- Middle trace (orange): HandSensor total acceleration vector (g) with current shift applied.
- Bottom trace (yellow): FingerSensor total acceleration vector (g) with current shift applied.

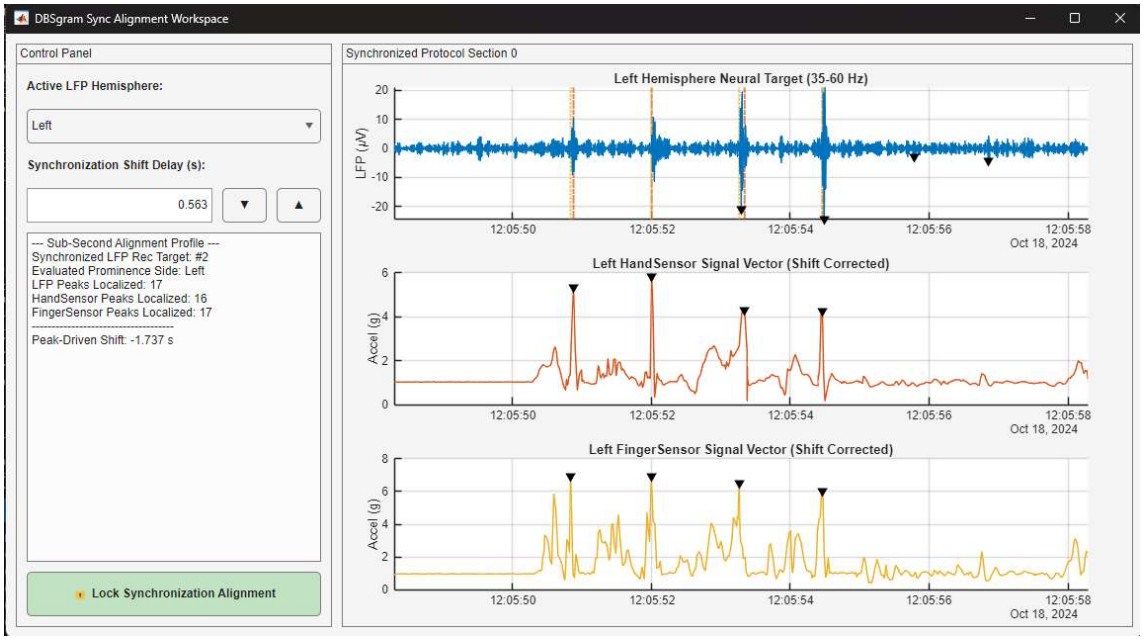


Figure 6 - Sync Alignment Workspace displaying LFP artifact alignment against kinematic taps

To review and adjust alignment, the user should first observe the information on the localized peak counts and an initial *Peak-Driven Shift* on the control panel. Then, the user can interact with the **Active LFP Hemisphere** dropdown selection and try to understand in what hemisphere the peaks have a higher signal to noise ratio. Finally, the user can interact with the increment (**Δ**) and decrement (**∇**) steppers to manually adjust the **Synchronization Shift Delay** (s) field. It's also possible for the user to set a certain value by clicking the input box and writing the amount through the keyboard. Once the peaks in the 3 streams are aligned, the user should click on **Lock Synchronization Alignment.** In result, the master GUI looks like Fig. 7 with all buttons in the Panel 1 in green and the **Audit Channel Data Inclusion** and **Remove Cardiac ECG SVD artifacts** buttons become enabled.

Note: After pressing the alignment button, data ingestion readjustments can only be done by closing the app and re-starting the toolbox. A warning dialog appears to alert users of this important detail.

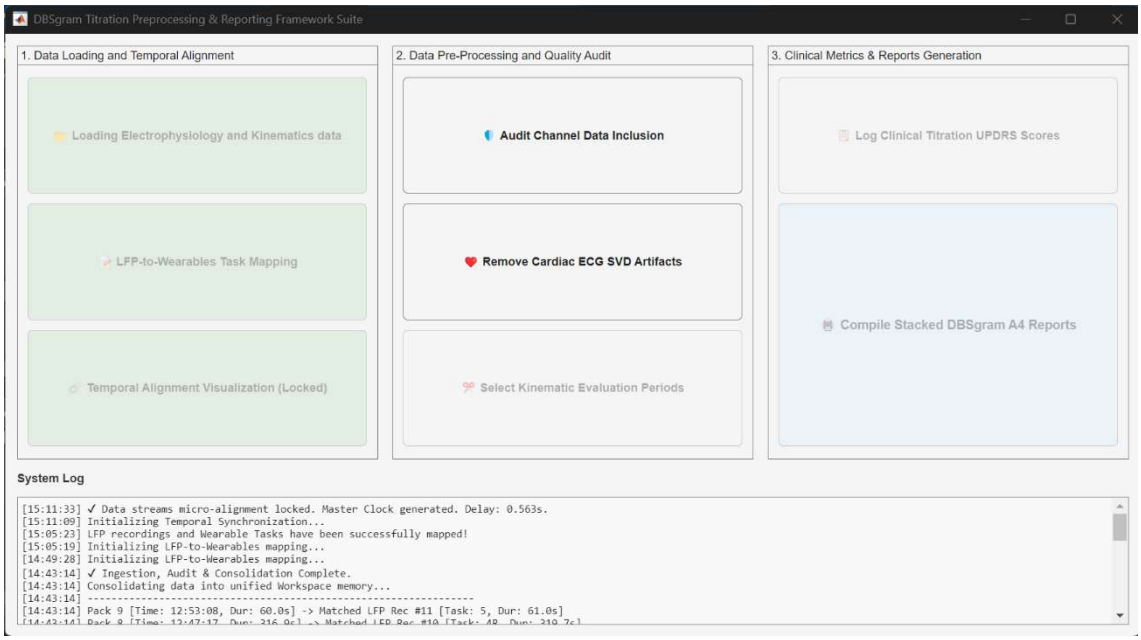


Figure 7 - Main Framework Suite with Phase 1 locked and Quality Audit options enabled

4) Multi-Modal Data Quality Audit

To prevent corrupted data from propagating into downstream metrics, both streams must pass a two-part interactive quality audit. A recording block is only included in final clinical analytics if it is explicitly verified and approved across both audit stages.

4.1) Electrophysiological Spectra Audit

After clicking on the **Audit Channel Data Inclusion** button on the main dashboard, a *Clinical Parallel Multi-Modal Quality Audit Workspace (Part 1)* window opens, which allows the user to cycle sequentially through every mapped protocol block.

Here, the user can see in a 2x2 grid plot, the data for each hemisphere side (when applicable). In particular, on the upper axis, is presented the LFP (µV, blue for the left side or green for the right side) overlaid directly against the stimulation amplitude (mA, orange) and, on the lower axis, a Multitaper Spectrogram (dB power/frequency), displaying 0-125 Hz across elapsed time. Audited channels display spectral power, and non-audited channels are masked in grey with an N/A marker in the title (see Fig. 8).

Channel viability for downstream analysis must be audit on this window and the user must confirm:

- LFP trace demonstrates physiological dynamics without saturation or rail-to-rail amplifier clipping
- LFP traces shouldn't show sustained telemetry dropout
- LFP traces shouldn't display stimulation artifacts contaminating the beta-band spectral.

If viability is confirmed check the box in **Include LEFT/RIGHT Hemisphere for Downstream Analytics**. If left unchecked, this block's LFP will be excluded from the final DBSgram Beta power calculations.

Click **Confirm & Next Section** to advance. Repeat for all blocks in the queue (e.g. Blocks 1 through 8).

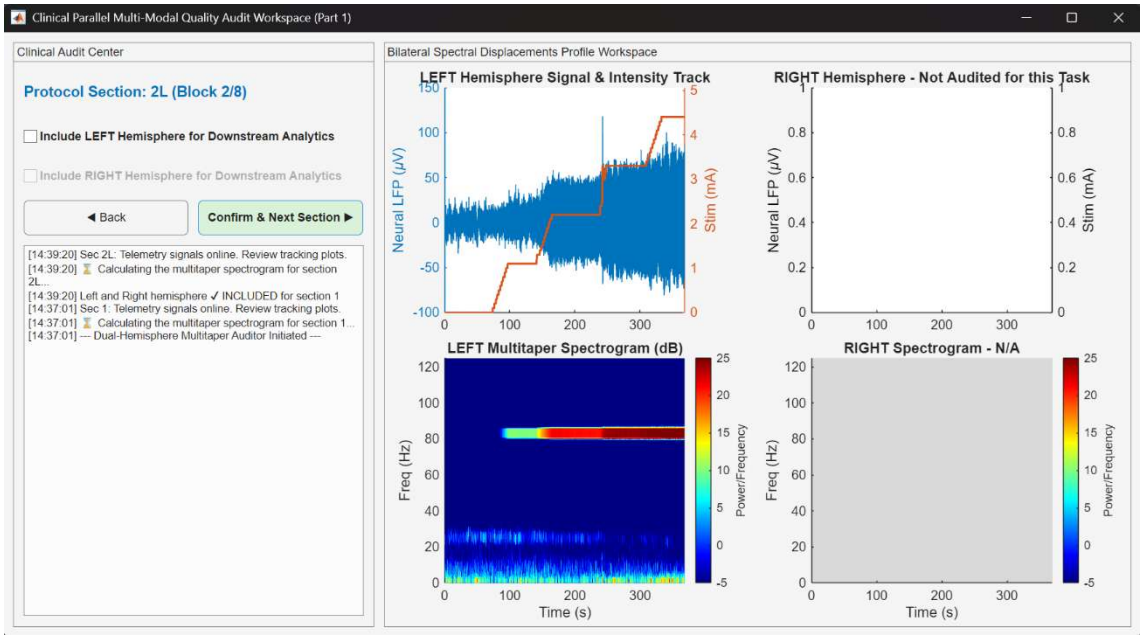


Figure 8 - Part 1 Quality Audit Workspace displaying LFP time-series and multitaper spectrogram

4.2) Cross-Modality Kinematic Verification

Once Part 1 finishes its final block, the interface automatically closes and reopens into *Multi-Modal Data Quality & Cross-Alignment Monitor (Part 2)*. This interface is a 2x3 grid with Left Hemisphere + Right Hemibody and Right Hemisphere + Left Hemibody columns. The traces are as follows:

- Top traces (blue/green): LFP signals (μV) overlaid with continuous stimulation current (mA).
- Middle traces: HandSensor total acceleration (g) for the specific hemibody.
- Bottom traces: FingerSensor total acceleration in g for the specific hemibody.

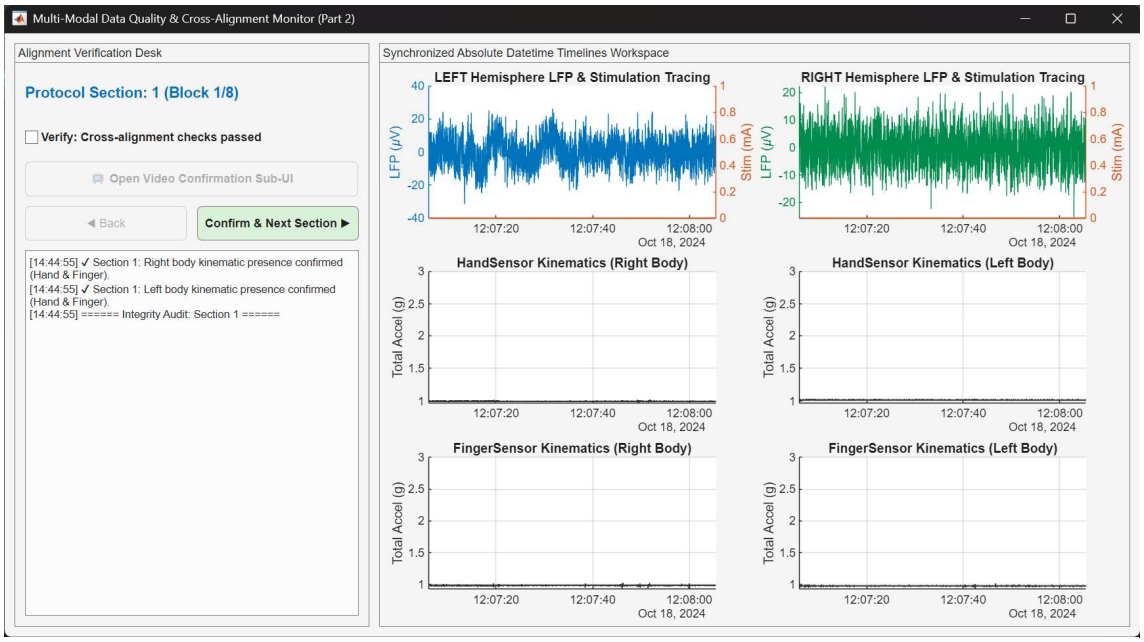


Figure 9 - Part 2 Cross-Alignment Monitor verifying bilateral LFP and wearable acceleration streams

The goal with this dashboard is to verify that kinematic bursts align with the expected clinical testing periods. If behavioral ambiguity exists, the recommendation is to inspect the respective video streams.

Similarly to the previous window, if the kinematic data is complete and temporally coherent with the stimulation block, check the **Verify: Cross alignment checks passed** box. After this click **Confirm & Next Section** Button, to validate effectively the checkbox. Repeat until all blocks have been audited.

After verifying the final block in Part 2, the audit window automatically saves the inclusion logical vector into the primary application workspace and closes.

The main UI is now fully prepared for **Cardiac ECG SVD Artifact Removal**, **Kinematic Evaluation Period Selection** and **Log Clinical Titration UPDRS scores** and will display as shown in Fig. 10.

Note: the user can choose to produce first clean version of the LFP by removing the ECG artifacts before doing the Data Audit part described in this section. Similarly, the Kinematic Evaluation can be performed before ECG removal as well. The order presented here was selected just because of button layout on the Panel 2 of the main UI.

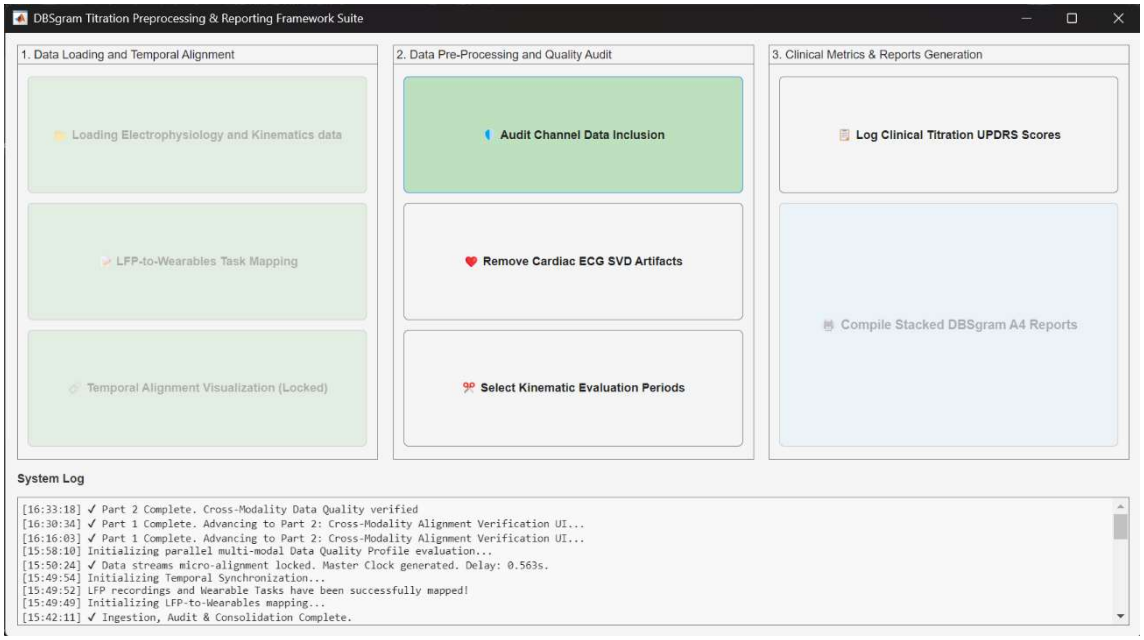


Figure 10 - Main Framework Suite after successful completion of the multi-modal quality audit

5) ECG Artifact Suppression

Implanted Pulse Generators (IPGs) positioned in the pectoral pocket frequently capture far-field electrocardiographic (ECG) volume conduction. These cardiac artifacts bleed across electrode channels, contaminating the LFP time-series. The toolbox provides a subspace Singular Value Decomposition (SVD) module to remove ECG components without eroding intrinsic neural rhythms.

On the main dashboard, click **Remove Cardiac ECG SVD artifacts** in Panel 2. Two coordinate windows will open simultaneously:

- *SVD Subspace Matrix Component Diagnostics* (Fig. 11)
  - Peri-Peak Template P-QRS-T Profile (Top-Left): Displays the signal-averaged cardiac template aligned to detected R-peaks (-0.2s to + 0.4s). A sharp, classical QRS deflection confirms cardiac volume conduction.
  - Singular Values Variance Spectrum (Top-Right): Ranks SVD components by explained variance (%). Dominant components capturing > 10-15% of total energy typically reflect heartbeat artifacts.
  - Orthogonal Singular U Vector Waves (Bottom): Displays the morphological waveforms of the top spatial components (Component 1 to Component 4). Components showing sharp, biphasic/triphasic QRS complexes indicate cardiac components suitable for rejection.

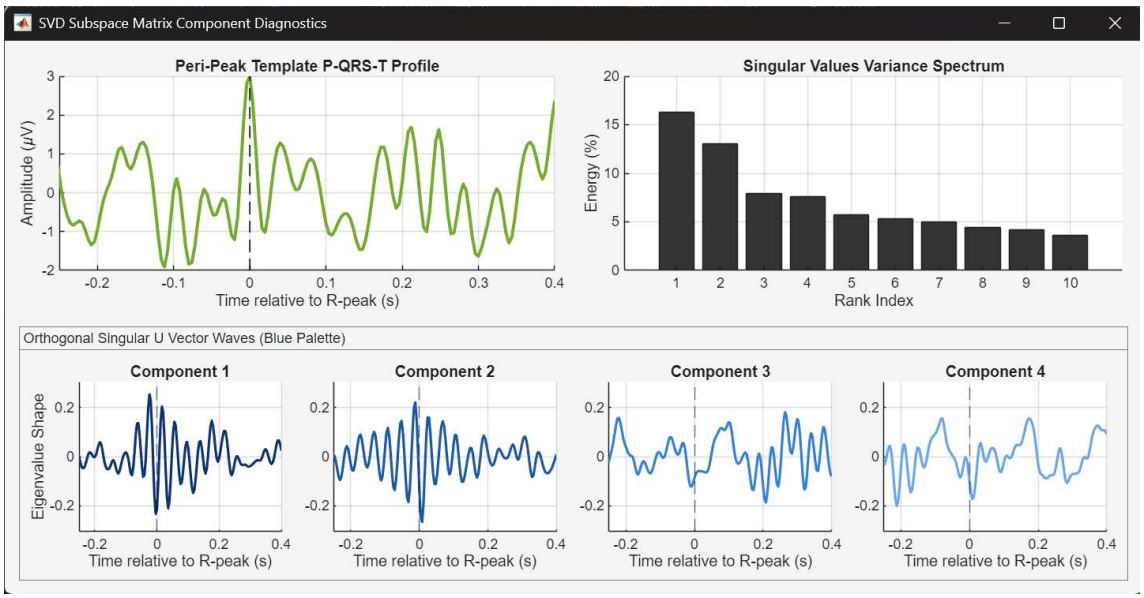


Figure 11 - SVD Subspace Matrix Component Diagnostics Interface

- *Interactive SVD Cardiac Artifact Suppression* (Fig. 12)

This window can be interacted with in terms of:

- **Protocol Target Step:** This dropdown allows the user to choose the desired recording section.
- **Target Evaluation Hemisphere:** The user can toggle between Left and Right hemisphere channels
- **Select SVD Component to Reject:** This dropdown allows the user to select a target component (e.g. Component 1) and click the **+** to remove it.
- Rejected Components: Label storing all the excluded components.

The graphs in the center and right part of the window containing on the top an overlay of the raw LFP trace (blue) and the reconstructed, cleaned LFP trace (dashed purple) against the stimulation amplitude (orange) and a Multitaper Spectrogram in the bottom, displaying the spectral power of the filtered signal only. Here, the user

should verify that periodic vertical striping (cardiac broadband artifacts) is eliminated without creating unnatural spectral troughs in the Beta band.

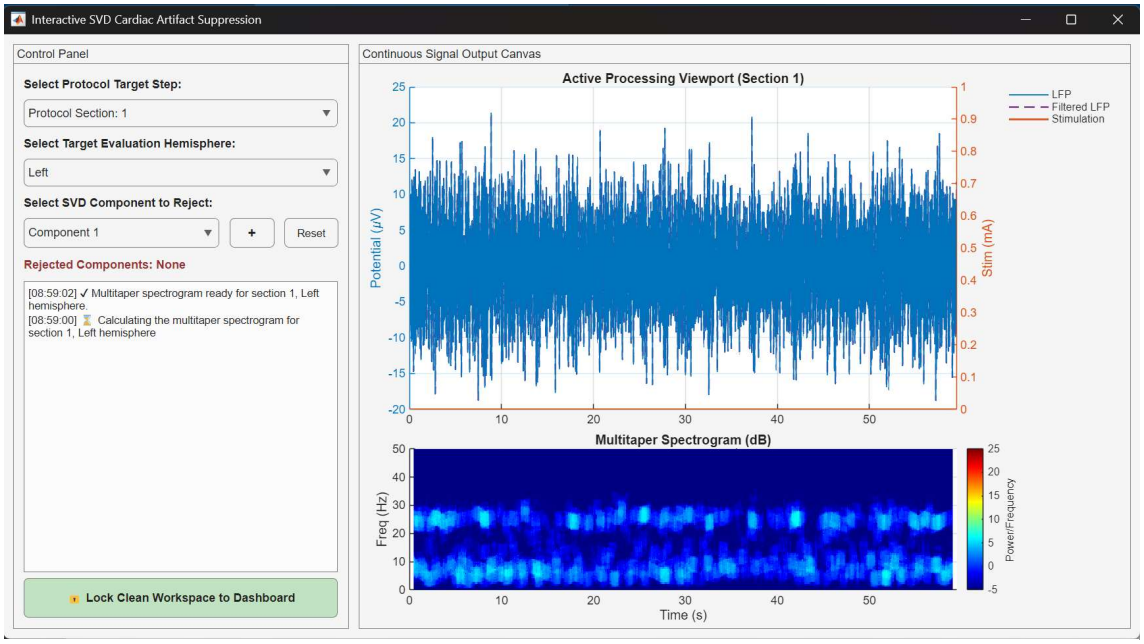


Figure 12 - Interactive SVD Cardiac Artifact Suppression Interface

If there ever happens to be an over-aggressive component rejection that removed physiological neural oscillations, click **Reset** to return to no components removed baseline.

Once filtering is satisfactory across all affected blocks, click **Lock Clean Workspace to Dashboard** to close both windows and return to the main UI. As shown in Fig. 13, the corresponding ECG artifact removal button changes to green.

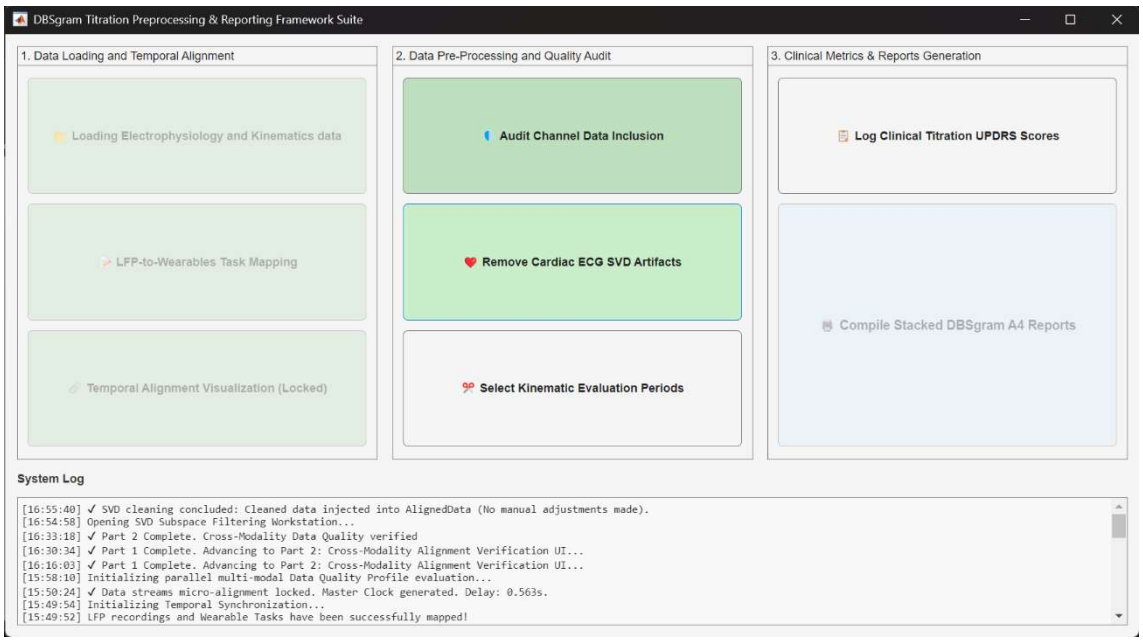


Figure 13 - Main Framework Suite with SVD Cleaning locked and Kinematic Selection enabled

6) Kinematic Window Bounding

Click in the **Select Kinematic Evaluation Periods** to open a new interface called *Specialized Synchronized Multilevel Kinematic Selector* (Fig. 14) to define precise temporal boundaries during which the patient performed the task at each titration level. The window presents the total acceleration (g) plooted alongside the continuous stimulation current (mA) for the HandSensor and FingerSensor on the top and bottom graph, respectively. As one can observe in the above picture, kinematic evaluation periods appear as distinct acceleration bursts during each stable stimulation plateau. The axes are time-aligned and linked and the vertical markers as well.

The automated boundary engine detects these intervals on launch. If necessary, click and drag the dashed boundary markers horizontally to align the window with the kinematic burst. If you need to restore the automated window bounds across all levels, click **Auto-Detect Boundaries.**

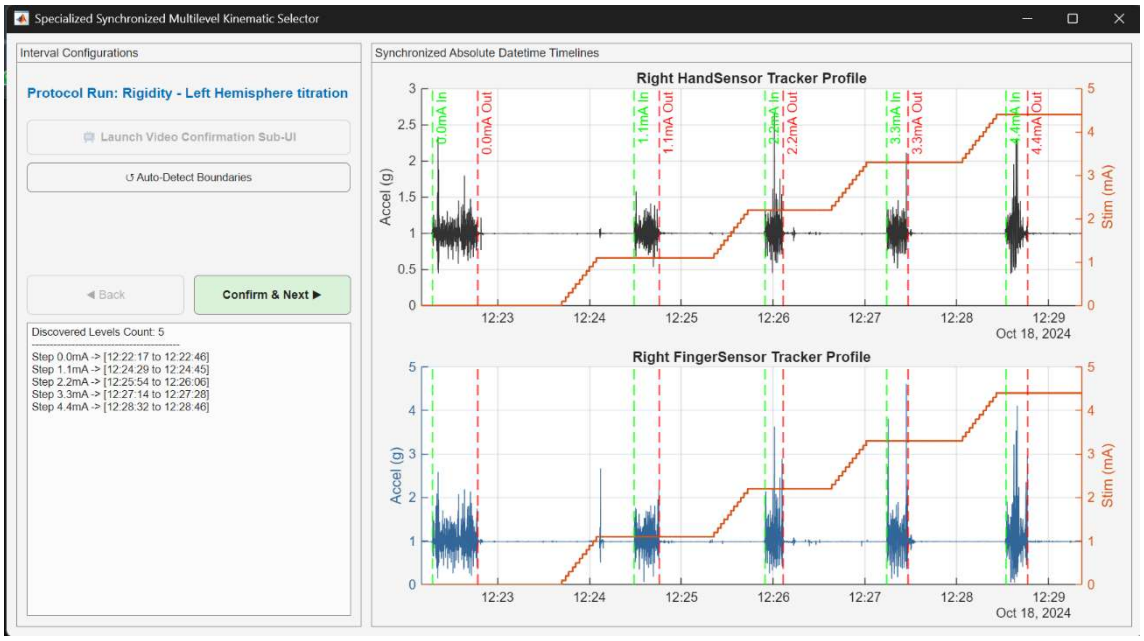


Figure 14 - Multilevel Kinematic Selector window bounding stage showing stimulation titration steps

Note: If the clinician needs to re-test a patient at a single stimulation step, multiple bursts may appear on that plateau. Always position the boundary markers around the final completed burst, as this reflects the steady-state clinical score logged by the neurologist

Once boundaries are set, click **Confirm & Next** button to proceed to cycle verification for this motor task.

7) Kinematic Cycle Detection & Peak Validation

After window bounding, the interface transitions into the cycle-level inspection view. This view extracts individual movement sweeps to compute biomechanical metrics. Details of how for both active motor tasks this view appears are presented.

To advance between protocols, click **Confirm Peaks & Next**. The suite will cycle sequentially through all remaining mapped kinematic blocks (e.g., contralateral Rigidity, contralateral Bradykinesia). On the second evaluation screen of the final kinematic protocol, the advance button dynamically relabels to **Finalize Kinematics**. Click this button to save all validated cycle matrices to workspace memory and return to the main dashboard.

7.1) Wrist Rigidity Protocol

For rigidity tasks, the engine separates passive wrist sweeps from activated maneuvers (Froment's sign activation via contralateral limb movement).

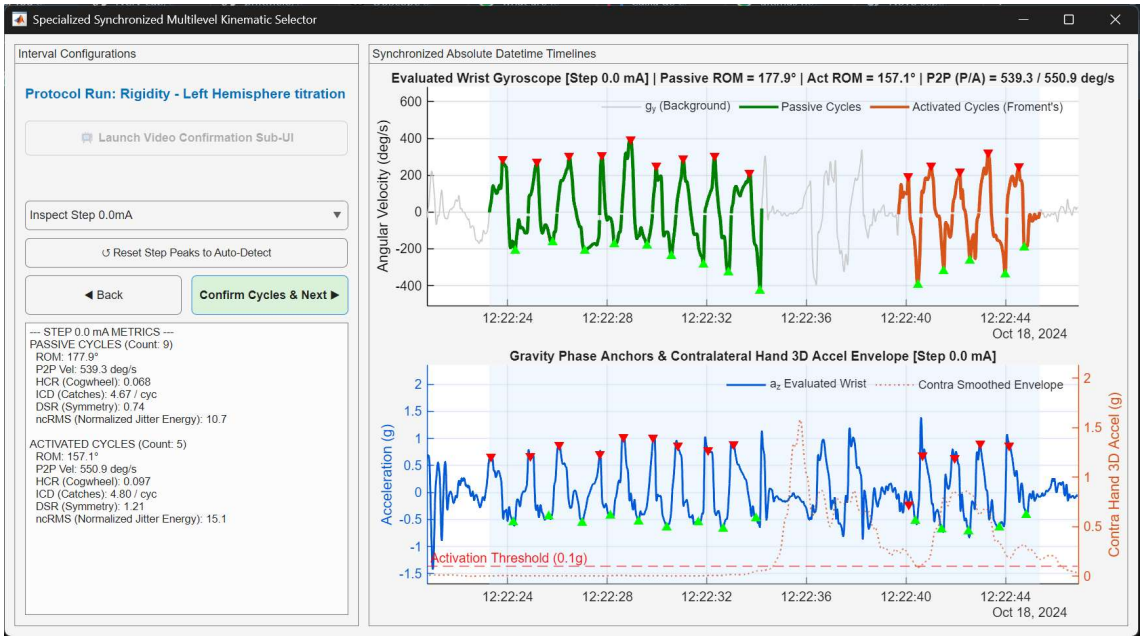


Figure 15 - Rigidity cycle detection display showing passive (green) and activated (orange) sweeps

Here (see example in Fig.15) two plots are presented:

- Upper Plot (Evaluated Wrist Gyroscope): Displays angular velocity (g_y, deg/s). Identified cycles are shaded in Green (Passive Swings) and Orange (Activated Swings). Red inverted triangles indicate flexion peaks and light green triangles mark extension troughs.

- Lower Plot (Gravity Anchors & Acceleration Envelope): Displays the wrist gravity vector (a_z, blue) alongside the smoothed 3D acceleration envelope of the contralateral hand (orange dashed). When contralateral activity crosses the Activation Threshold (0.1g), sweeps are automatically classified as Activated.

Cycle-by-cycle metrics are logged in the left panel: Range of Motion (ROM), Peak-to-Peak Velocity (P2P Vel), Harmonic Cogwheel Ratio (HCR), Inter-Catch Distance (ICD), Directional Symmetry Ratio (DSR), and normalized Jitter Energy (ncRMS). To switch between current steps using the **Inspect Step** dropdown.

Due to erratic tremors or cogwheel catches, automated peak detection may occasionally register an incomplete half-sweep or motion transient (e.g., the premature activated cycle visible at 12:22:40 in Fig. 15). In these cases, click directly on the false peak marker on the top plot to delete it. The engine will discard the invalid cycle and immediately recompute the step's summary metrics.

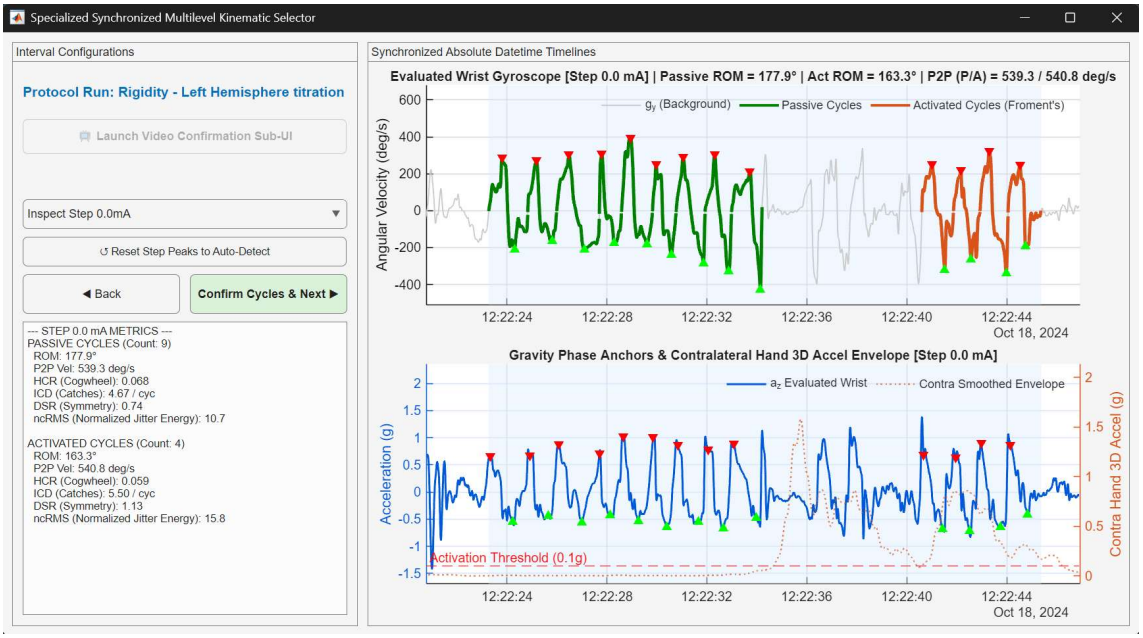


Figure 16 - Rigidity cycle interface after manual removal of a false cycle at 12:22:40

Note: The **Reset Step Peaks to Auto-Detect** button recalculates and resets peak detections globally across all stimulation steps for this protocol, overwriting any manual edits. Use it with due caution!

7.2) Bradykinesia Protocol

When processing hand open-close tasks for bradykinesia assessments, the cycle engine measures movement amplitude decay and velocity fatigue, as demonstrated in Fig. 17.

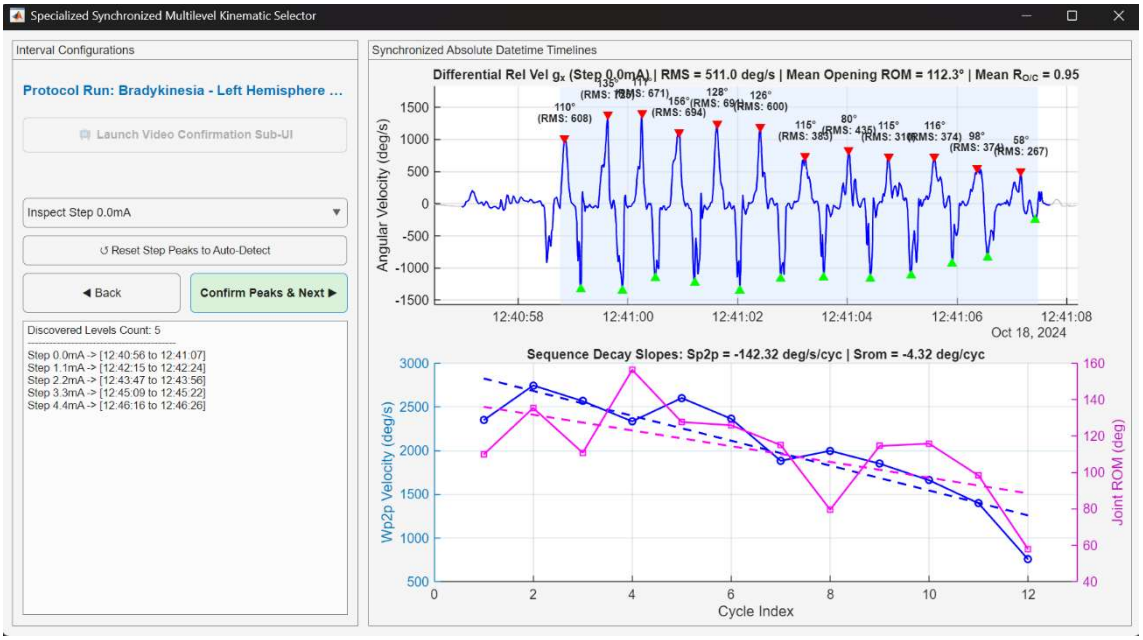


Figure 17 - Bradykinesia kinematic evaluation interface showing velocity traces and sequence decay slopes

Here two plots are presented:

- Upper Plot (Differential Relative Velocity): Displays angular velocity (g_x, deg/s) with individual opening peaks annotated with local opening ROM (°) and cycle RMS. The title displays summary metrics: global RMS velocity, Mean Opening ROM, and Opening-to-Closing Ratio ($R_{O/C}$).

- Lower Plot (Sequence Decay Slopes): Plots Peak-to-Peak Velocity (blue trace, left axis) and Joint ROM (magenta trace, right axis) across sequential cycle indices. Dashed regression trendlines calculate the sequence decay slopes (S_{p2p} and S_{rom}), quantifying motor decrement and physical exhaustion over time.

Similarly to what should be done for rigidity, inspect every step using the **Inspect Step** dropdown. Remove spurious cycles or half-swings by clicking their peak markers.

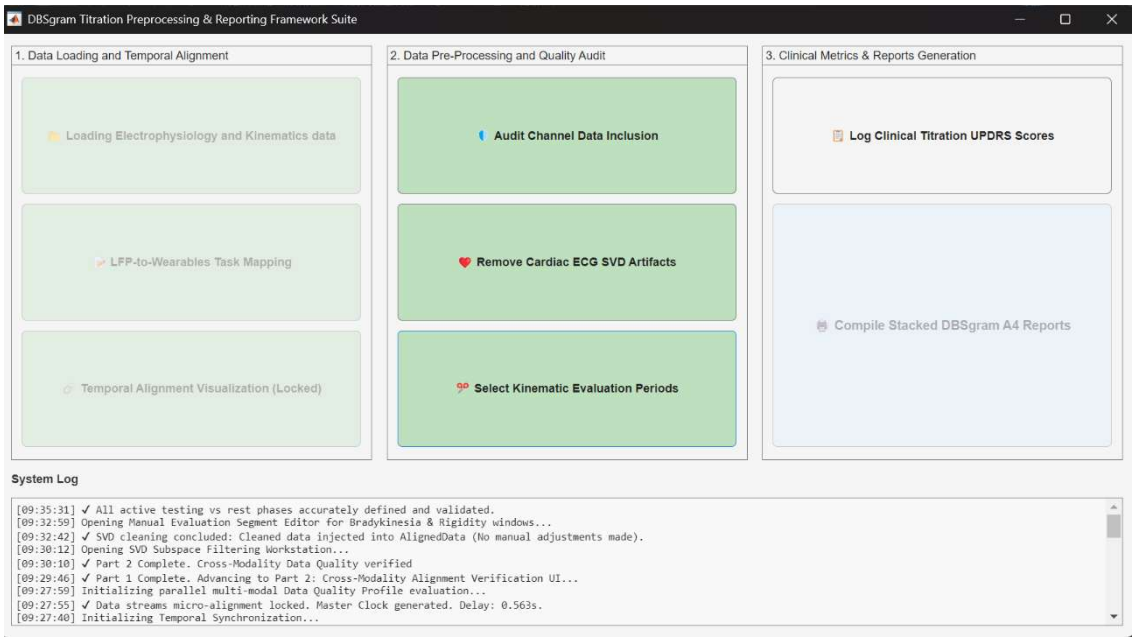


Figure 18 - Main Framework Suite with Kinematics complete and Clinical Scoring unlocked

8) Clinical Titration UPDRS Score Logging

To enable multimodal correlation, clinical scoring (UPDRS) evaluations logged during the patient's in-clinic titration must be paired with the objective sensor datasets. Click on **Log Clinical Titration UPDRS Scores** on the main dashboard to open *Multi-Symptom Titration Workflow Clinical Overview Panel* (see Fig. 19). If a UPDRS.xlsx file with the specific format mentioned in the first section of this guide exists in the data ingestion folder, the values are preallocated.

The interface cycles through symptoms sequentially. First, it asks for the information for each titrated step for Left Hemisphere / Right Hemibody than switches to their contralaterals. It displays the respective current amplitude steps and requests the corresponding UPDRS Motor Score in the dropdown.

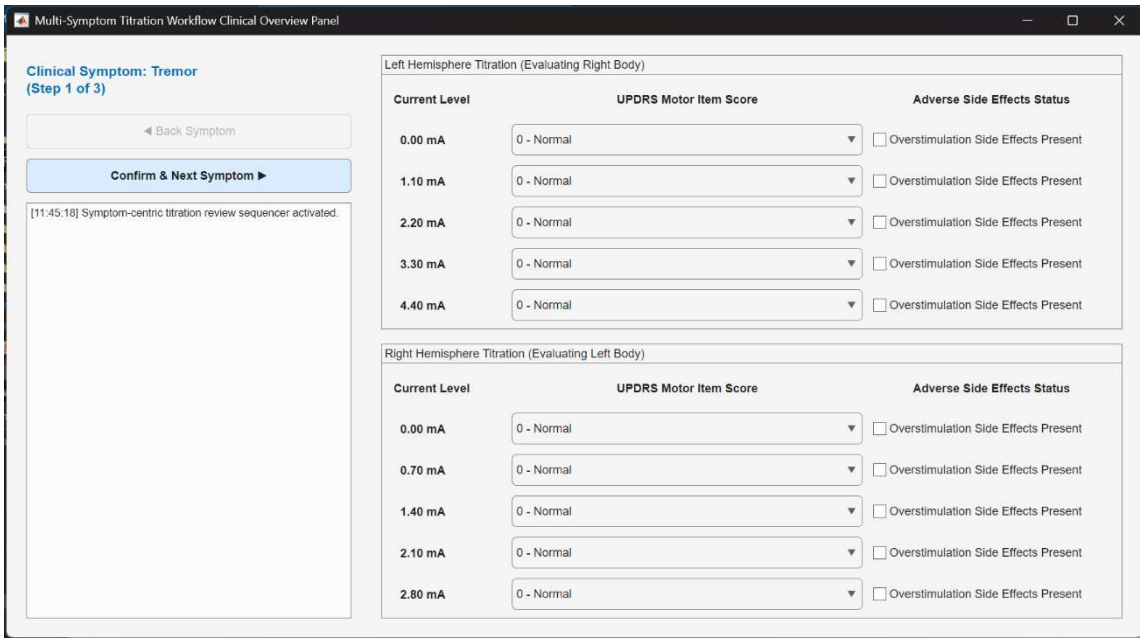


Figure 19 - Multi-Symptom Titration Workflow Clinical Overview Panel for UPDRS logging

If an amplitude step triggered capsular side effects, paresthesias, dysarthria, or motor contractions, check the **Overstimulation Side Effects Present** box. This boundary marker defines the upper limit of the therapeutic window on the final DBSgram report.

Click **Confirm & Next Symptom** to cycle through all motor domains. On the final symptom stage, click **Finalize Data Blueprint to Dashboard.** Clicking this button will score every profile in the workspace, closing the UI and returning the user to the main UI, which has now the DBSgram Report engine enabled (see Fig. 20).

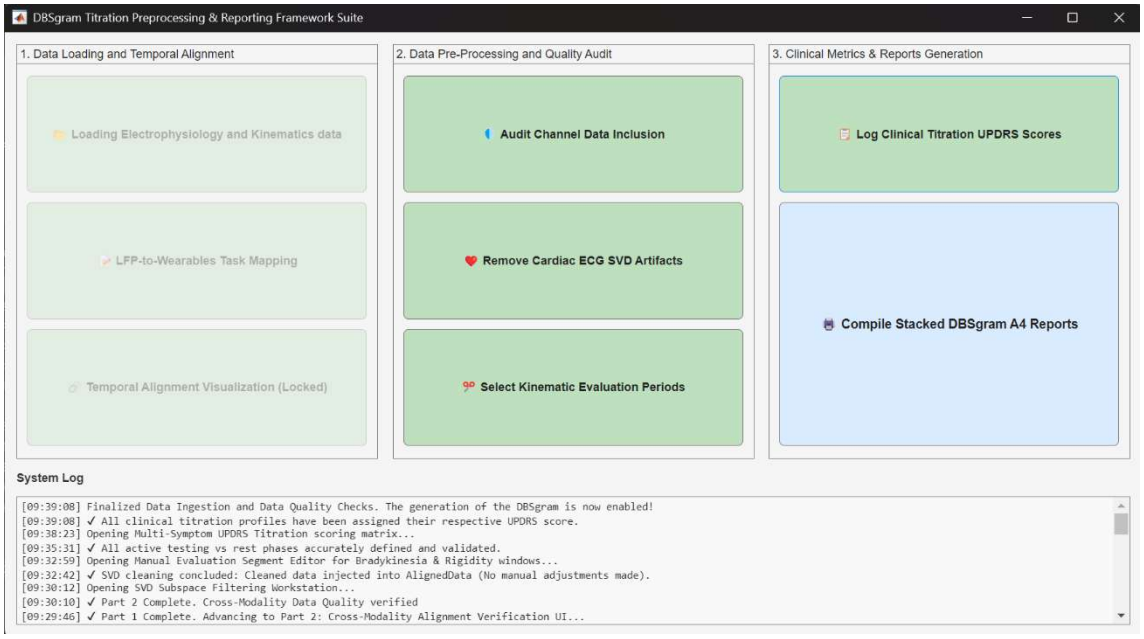


Figure 20 - Main Framework Suite fully configured and ready for DBSgram Report generation